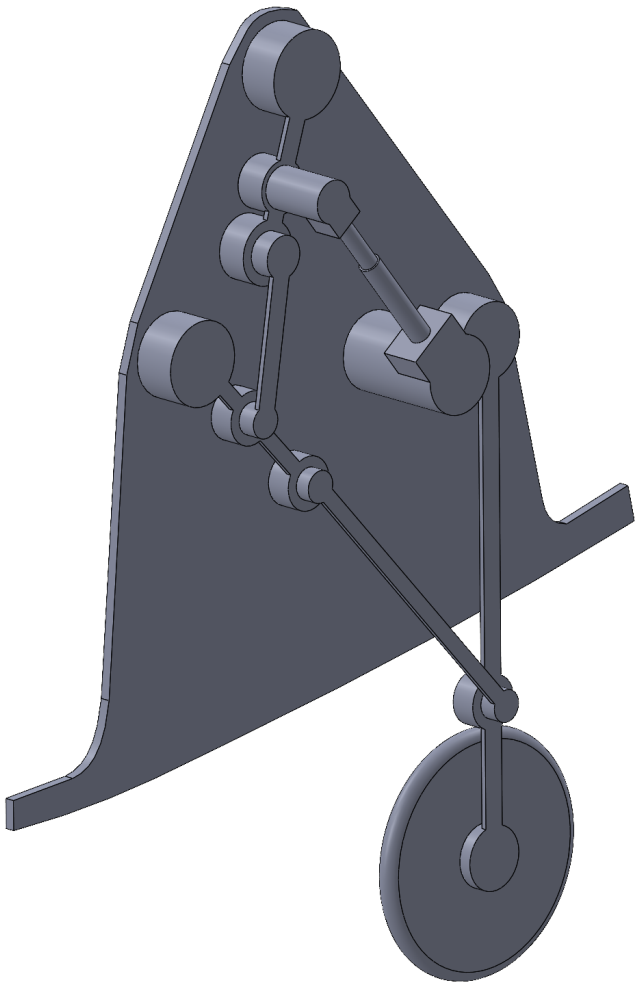
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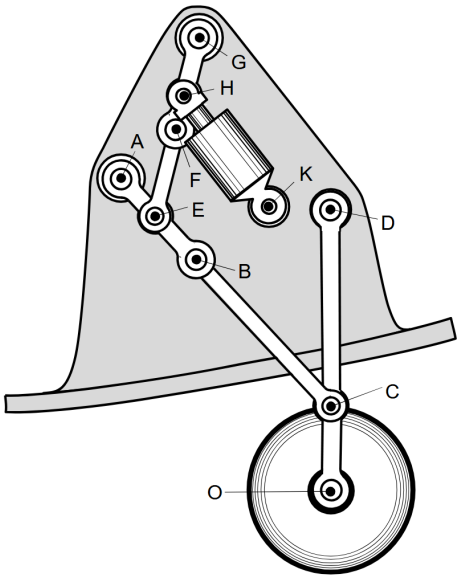
Dinâmica de Sistemas Mecânicos

Homework 1 – Plane Landing Gear

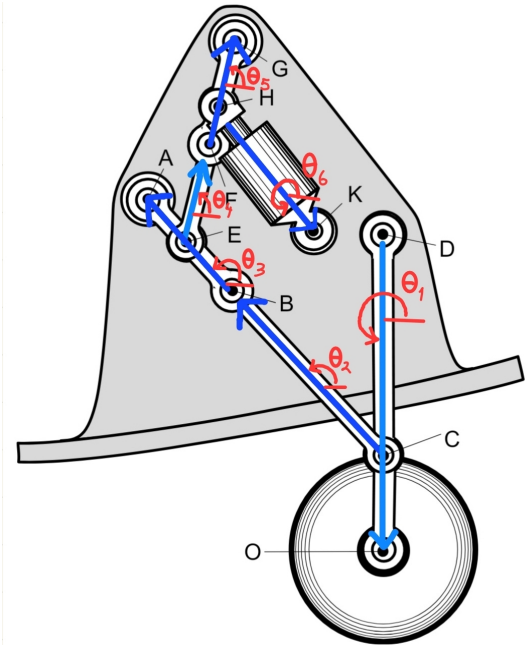
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1 Mechanism – Landing Gear I



Landing Gear schematic



Template for Mathematical Model

1.1 Links Measures (in meters)

$$
\begin{aligned}
&\mathrm{l}_{AE} = 0.09, \quad \mathrm{l}_{EB} = 0.097, \quad \mathrm{l}_{AEB} = \mathrm{l}_{AE} + \mathrm{l}_{EB} = 0.187,\\
&\mathrm{l}_{BC} = 0.335,\\
&\mathrm{l}_{DC} = 0.331, \quad \mathrm{l}_{CO} = 0.143, \quad \mathrm{l}_{DCO} = \mathrm{l}_{DC} + \mathrm{l}_{CO} = 0.474,\\
&\mathrm{l}_{EF} = 0.148, \quad \mathrm{l}_{FH} = 0.06, \quad \mathrm{l}_{HG} = 0.101, \quad \mathrm{l}_{FHG} = \mathrm{l}_{FH} + \mathrm{l}_{HG} = 0.161,\\
&\mathrm{l}_{GAx} = 0.132, \quad \mathrm{l}_{GAy} = 0.237,\\
&\mathrm{l}_{GKx} = 0.116, \quad \mathrm{l}_{GKy} = 0.285,\\
&\mathrm{l}_{GDx} = 0.221, \quad \mathrm{l}_{GDy} = 0.291,\\
&\mathrm{l}_{DKx} = 0.105, \quad \mathrm{l}_{DKy} = 0.005.
\end{aligned}
$$

1.2 Driver – Linear actuator – Hydraulic cylinder

$$\mathrm{l}_{HK}^{\min} = 0.233m \quad \mathrm{l}_{HK}^{\max} = 0.333m \quad \dot{\theta} = \frac{2\pi}{10} rad/s \quad N_{\text{Coordinates}} = 7$$

1.3 Initial angles with respect to the horizontal

$$
\begin{aligned}
&\theta_1 = 270^\circ \quad \theta_2 = 132.57^\circ \quad \theta_3 = 132.42^\circ\\
&\theta_4 = 75.93^\circ \quad \theta_5 = 76.29^\circ \quad \theta_6 = 306.95^\circ\\
&l_{HK} = \mathrm{l}_{HK}^{\min}
\end{aligned}
$$

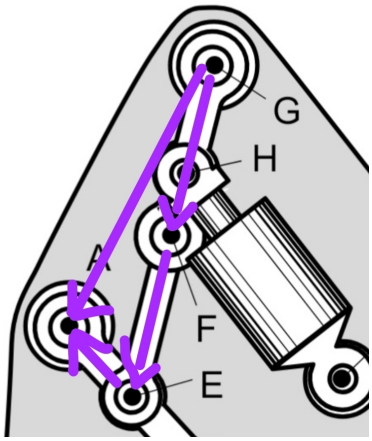
1.4 Newton-Raphson Method and time parameters

$$
\begin{aligned}
&\text{Nonlinear tolerance: } 10^{-6} \quad \text{Maximum number of iterations: } 20\\
&t_{\text{start}} = 0.0s \quad t_{\text{end}} = 10.0s \quad \Delta t = 0.04s
\end{aligned}
$$

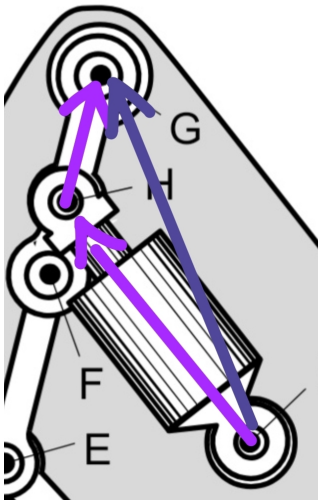
$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \\ \theta_6 \\ l_{HK} \end{bmatrix}$$

With all the parameters defined above the mathematical model can be defined by using vectors relationships. It is expected to have 7 equations in order to solve and find all 7 degrees of freedom. Having 3 vector loops means having 6 equations, the last one is defined with the driver actuator equation.

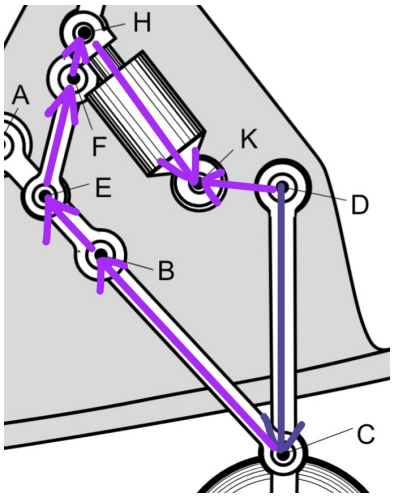
1.5 Vector Loops



Vector Loop (1)



Vector Loop (2)



Vector Loop (3)

Comparing the vectors of each vector loop with the Template the relations can be defined:

$$\vec{l}_{FHG} + \vec{l}_{FE} + \vec{l}_{EA} - \vec{l}_{AG} = 0 \tag{1}$$

$$\vec{l}_{KH} + \vec{l}_{HG} - \vec{l}_{KG} = 0 \tag{2}$$

$$\vec{l}_{DC} + \vec{l}_{CB} + \vec{l}_{BE} + \vec{l}_{EF} + \vec{l}_{HK} - \vec{l}_{DK} = 0 \tag{3}$$

$$\mathbf{\Phi}_{vector-loops}(q,t) = 0 = \begin{bmatrix} l_{DC}\cos\theta_1 + l_{BC}\cos\theta_2 + l_{EB}\cos\theta_3 + l_{EF}\cos\theta_4 + l_{FH}\cos\theta_5 + l_{HK}\cos\theta_6 + l_{DKx} \\ l_{DC}\sin\theta_1 + l_{BC}\sin\theta_2 + l_{EB}\sin\theta_3 + l_{EF}\sin\theta_4 + l_{FH}\sin\theta_5 + l_{HK}\sin\theta_6 + l_{DKy} \\ -l_{FHG}\cos\theta_5 - l_{EF}\cos\theta_4 + l_{AE}\cos\theta_3 + l_{GAx} \\ -l_{FHG}\sin\theta_5 - l_{EF}\sin\theta_4 + l_{AE}\sin\theta_3 + l_{GAy} \\ l_{HG}\cos\theta_5 - l_{HK}\cos\theta_6 + l_{GKx} \\ l_{HG}\sin\theta_5 - l_{HK}\sin\theta_6 - l_{GKy} \end{bmatrix} \tag{4}$$

1.6 Driver Actuator Equation

In order to retract the wheel, the hydraulic cylinder adjusts its length by increasing it. The actuator length can be defined as:

$$l_{HK}(t) = l_{HK}^{\min} + \frac{1}{2}\left(l_{HK}^{\max} - l_{HK}^{\min}\right)\left[1 - \cos(\omega t)\right] \tag{5}$$

1.7 Mathematical Model

With all the kinematic relationships and vector loop equations defined, the mechanism can be fully described. This allows the calculation of key matrices, such as the Jacobian matrix, which are essential for solving position, velocity, and acceleration analysis of the system. The equations below can be easily obtained by just derivating in time the kinematic equations Φ , position equation.

$$\dot{\mathbf{\Phi}}(\mathbf{q}, t) \equiv \mathbf{J}\dot{\mathbf{q}} - \boldsymbol{\nu} = \mathbf{0} \tag{6}$$

$$\ddot{\mathbf{\Phi}}(\mathbf{q}, \dot{\mathbf{q}}, t) \equiv \frac{d}{dt}\left(\mathbf{J}\dot{\mathbf{q}}\right) - \frac{\partial \boldsymbol{\nu}}{\partial t} = \mathbf{0} \tag{7}$$

The goal is to solve this matrix equations in order to obtain the generalized coordinates $\mathbf{q}(t)$, velocities $\dot{\mathbf{q}}(t)$, and accelerations $\ddot{\mathbf{q}}(t)$.

Since the kinematic constraint equations, $\mathbf{\Phi}(\mathbf{q}, t)$, are nonlinear, an iterative method is required. A common and effective approach is the *Newton-Raphson* method.

Position equations:

$$\mathbf{\Phi}(q,t) = \begin{bmatrix} l_{DC}\cos\theta_1 + l_{BC}\cos\theta_2 + l_{EB}\cos\theta_3 + l_{EF}\cos\theta_4 + l_{FH}\cos\theta_5 + l_{HK}\cos\theta_6 + l_{DKx} \\ l_{DC}\sin\theta_1 + l_{BC}\sin\theta_2 + l_{EB}\sin\theta_3 + l_{EF}\sin\theta_4 + l_{FH}\sin\theta_5 + l_{HK}\sin\theta_6 + l_{DKy} \\ -l_{FHG}\cos\theta_5 - l_{EF}\cos\theta_4 + l_{AE}\cos\theta_3 + l_{GAx} \\ -l_{FHG}\sin\theta_5 - l_{EF}\sin\theta_4 + l_{AE}\sin\theta_3 + l_{GAy} \\ l_{HG}\cos\theta_5 - l_{HK}\cos\theta_6 + l_{GKx} \\ l_{HG}\sin\theta_5 - l_{HK}\sin\theta_6 - l_{GKy} \\ l_{HK} - \left(l_{HK}^{\min} + \frac{1}{2}(l_{HK}^{\max} - l_{HK}^{\min})\left[1 - \cos(\omega t)\right]\right) \end{bmatrix} \tag{8}$$

Jacobian:

$$\mathbf{J} = \begin{bmatrix} -l_{DC}\sin\theta_1 & -l_{BC}\sin\theta_2 & -l_{EB}\sin\theta_3 & -l_{EF}\sin\theta_4 & -l_{FH}\sin\theta_5 & -l_{HK}\sin\theta_6 & \cos\theta_6 \\ l_{DC}\cos\theta_1 & l_{BC}\cos\theta_2 & l_{EB}\cos\theta_3 & l_{EF}\cos\theta_4 & l_{FH}\cos\theta_5 & l_{HK}\cos\theta_6 & \sin\theta_6 \\ 0 & 0 & -l_{AE}\sin\theta_3 & l_{EF}\sin\theta_4 & l_{FHG}\sin\theta_5 & 0 & 0 \\ 0 & 0 & l_{AE}\cos\theta_3 & -l_{EF}\cos\theta_4 & -l_{FHG}\cos\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 0 & -l_{HG}\sin\theta_5 & l_{HK}\sin\theta_6 & -\cos\theta_6 \\ 0 & 0 & 0 & 0 & l_{HG}\cos\theta_5 & -l_{HK}\cos\theta_6 & -\sin\theta_6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{9}$$

Right Side of the velocity equation:

$$\boldsymbol{\nu} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\frac{1}{2}(l_{HK}^{\max} - l_{HK}^{\min})\,\omega\sin(\omega t) \end{bmatrix} \tag{10}$$

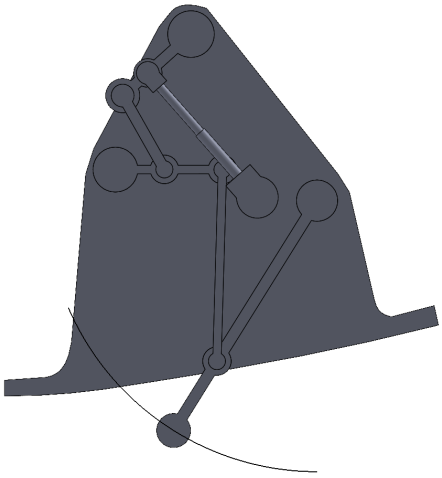
Right Side of the acceleration equation:

$$\boldsymbol{\gamma} = \begin{bmatrix} l_{DC}\cos\theta_1\,\dot{\theta}_1^2 + l_{BC}\cos\theta_2\,\dot{\theta}_2^2 + l_{EB}\cos\theta_3\,\dot{\theta}_3^2 + l_{EF}\cos\theta_4\,\dot{\theta}_4^2 + l_{FH}\cos\theta_5\,\dot{\theta}_5^2 + 2\dot{l}_{HK}\sin\theta_6\,\dot{\theta}_6 + l_{HK}\cos\theta_6\,\dot{\theta}_6^2 \\ l_{DC}\sin\theta_1\,\dot{\theta}_1^2 + l_{BC}\sin\theta_2\,\dot{\theta}_2^2 + l_{EB}\sin\theta_3\,\dot{\theta}_3^2 + l_{EF}\sin\theta_4\,\dot{\theta}_4^2 + l_{FH}\sin\theta_5\,\dot{\theta}_5^2 - 2\dot{l}_{HK}\cos\theta_6\,\dot{\theta}_6 + l_{HK}\sin\theta_6\,\dot{\theta}_6^2 \\ l_{AE}\cos\theta_3\,\dot{\theta}_3^2 - l_{EF}\cos\theta_4\,\dot{\theta}_4^2 - l_{FHG}\cos\theta_5\,\dot{\theta}_5^2 \\ l_{AE}\sin\theta_3\,\dot{\theta}_3^2 - l_{EF}\sin\theta_4\,\dot{\theta}_4^2 - l_{FHG}\sin\theta_5\,\dot{\theta}_5^2 \\ l_{HG}\cos\theta_5\,\dot{\theta}_5^2 - 2\dot{l}_{HK}\sin\theta_6\,\dot{\theta}_6 - l_{HK}\cos\theta_6\,\dot{\theta}_6^2 \\ l_{HG}\sin\theta_5\,\dot{\theta}_5^2 + 2\dot{l}_{HK}\cos\theta_6\,\dot{\theta}_6 - l_{HK}\sin\theta_6\,\dot{\theta}_6^2 \\ -\frac{1}{2}(l_{HK}^{\max} - l_{HK}^{\min})\,\omega^2\cos(\omega t) \end{bmatrix} \tag{11}$$

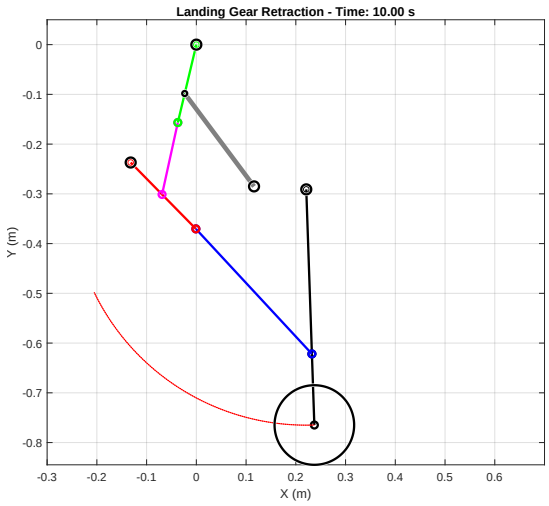
2 Solution analysis

In order to model the gear, tools such as SolidWorks and MATLAB were used. SolidWorks provided an intuitive way to visualize the mechanism without requiring any theoretical or mathematical equations, by modeling the problem using its own simulation engine. MATLAB, on the other hand, employed all the mathematical equations derived previously to perform a detailed kinematic and dynamic analysis.

2.1 Path Traced for point O (center of the wheel)

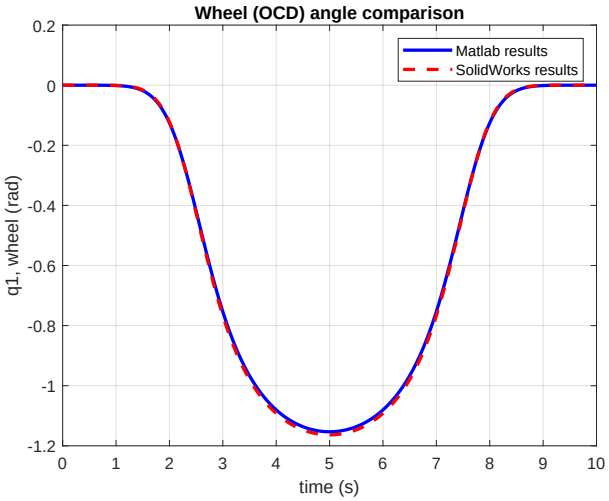


SolidWorks Path Drawned

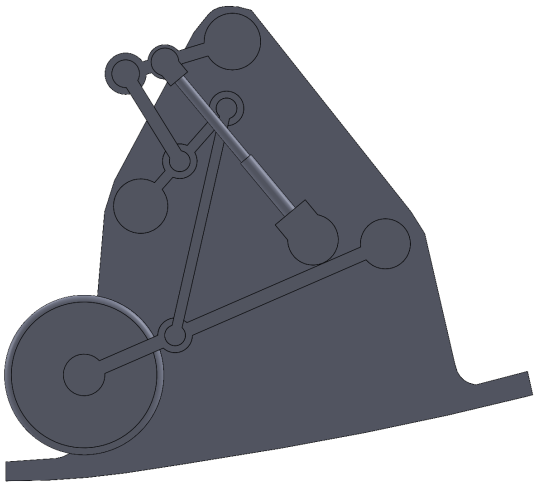


MATLAB Path Drawned

2.2 Comparison between both tools



Tool results comparison



Wheel fully retracted at t=5s

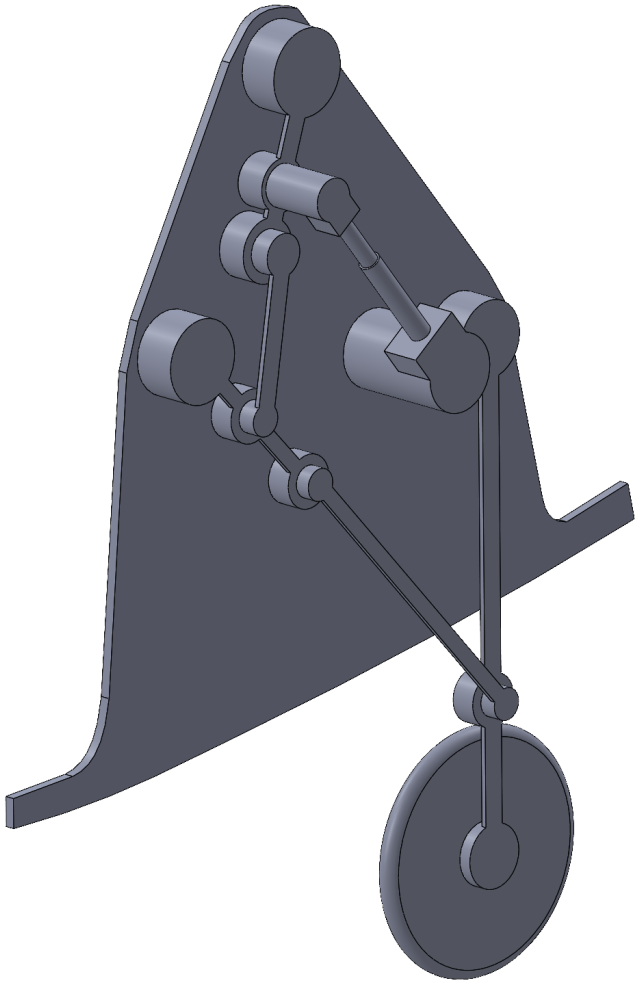
It can be concluded that both MATLAB and SolidWorks produced very similar results. By comparing the graphs, a discrepancy of approximately 1% can be observed between the two results (specially in the 5 second step). This occurs due to property approximations, in particular the link lengths. While modeling in SolidWorks, some lengths were defined visually in order to match the mechanism's geometry as depicted in the schematic. Another possible explanation for the discrepancy is the behavior of the SolidWorks simulation engine, which may apply its own internal approximations and constraints during the visualization and motion analysis.

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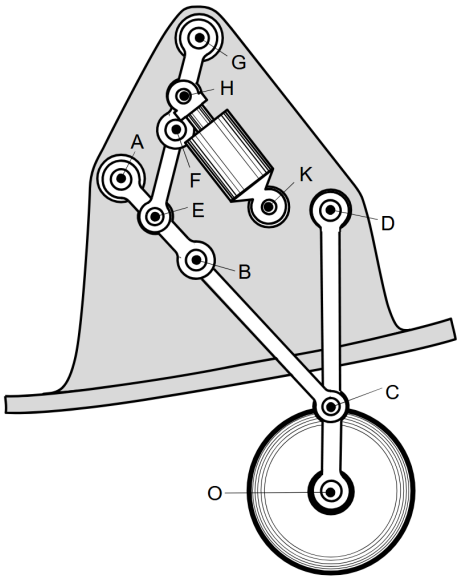
Dinâmica de Sistemas Mecânicos

Homework 2 – Plane Landing Gear with MUBOKAP

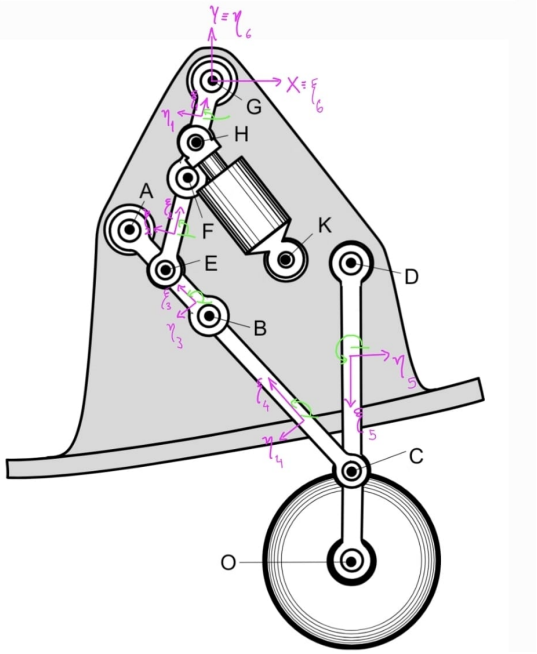
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1 Mechanism – Landing Gear I



Landing Gear schematic



Muti-Body Model

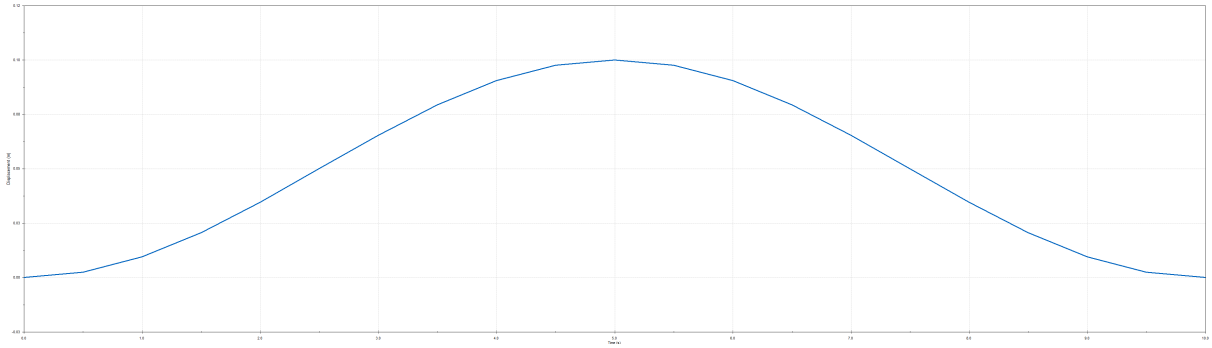
1.1 Body references, center of mass coordinates, and angles between the body axis and global axis

Bar Points	Body Number	Center of Mass Coordinates	Angle (rad)
FHG	1	(−0.010, −0.030)	1.33151
EF	2	(−0.050, −0.210)	1.32523
AEB	3	(−0.110, −0.260)	2.31116
BC	4	(0.110, −0.500)	2.31378
DCO	5	(0.22, −0.43)	4.71239
Base	6	(0.000, 0.000)	0.000

1.2 Connections between bodies (Joints)

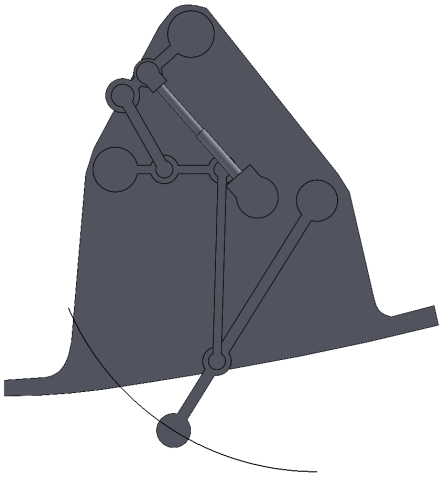
Bodies Connected	Coords locais do primeiro	Coords locais do segundo
6 1	(0.000, 0.000)	(0.030, 0.000)
6 5	(0.221, −0.291)	(−0.140, 0.000)
6 3	(−0.132, −0.237)	(0.030, 0.000)
1 2	(−0.130, 0.000)	(0.050, 0.000)
2 3	(−0.10, 0.000)	(−0.060, 0.000)
3 4	(−0.160, 0.000)	(0.170, 0.000)
4 5	(−0.160, 0.000)	(0.190, 0.000)

1.3 Driver Actuator Equation

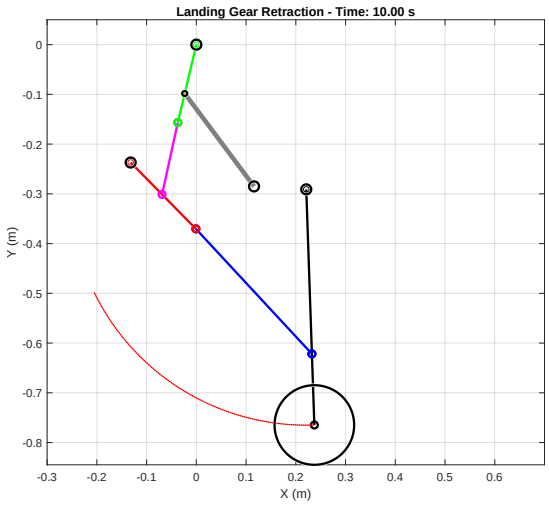


Warning: this is only the displacement equation. It is need to add the initial length for the Driver file on MUBOKAP.

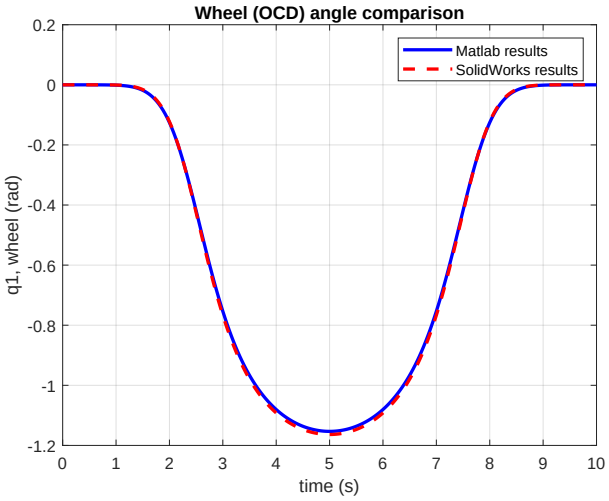
2 Solution analysis



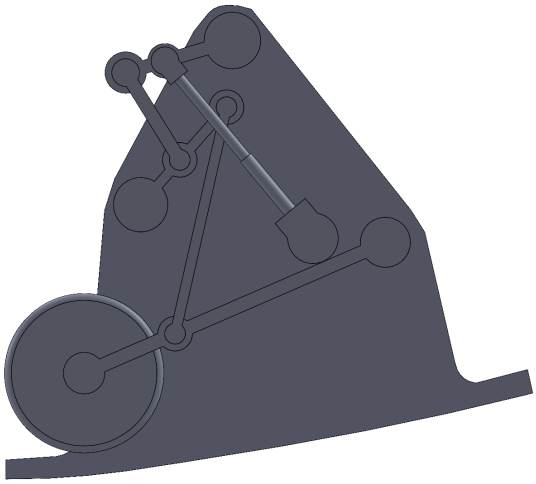
SolidWorks Path Drawned



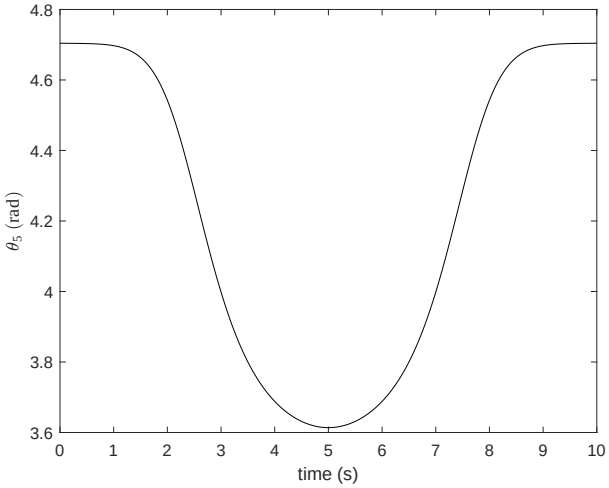
MATLAB Path Drawned



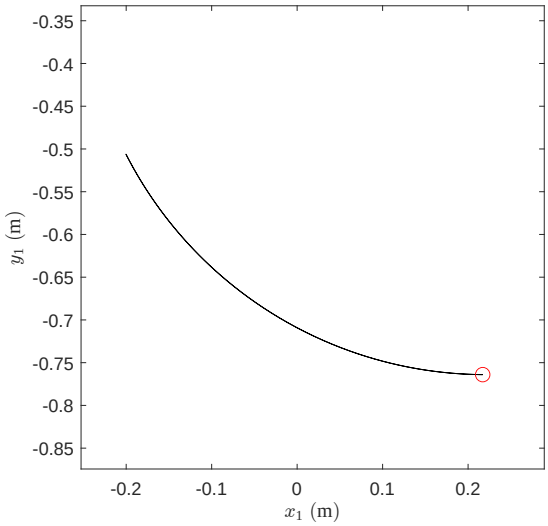
MATLAB and SolidWorks results comparison. q1 is the same as θ_5 for this Homework



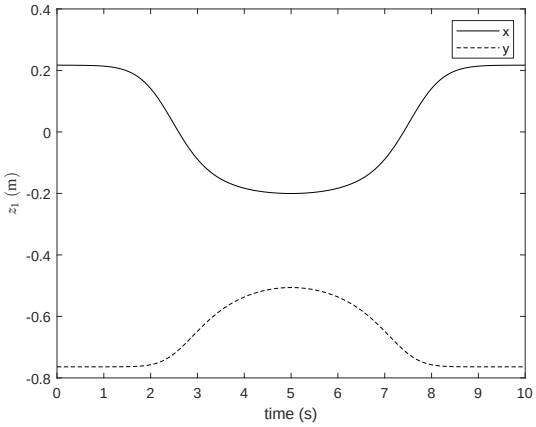
Wheel fully retracted at t=5s



MUBOKAP results



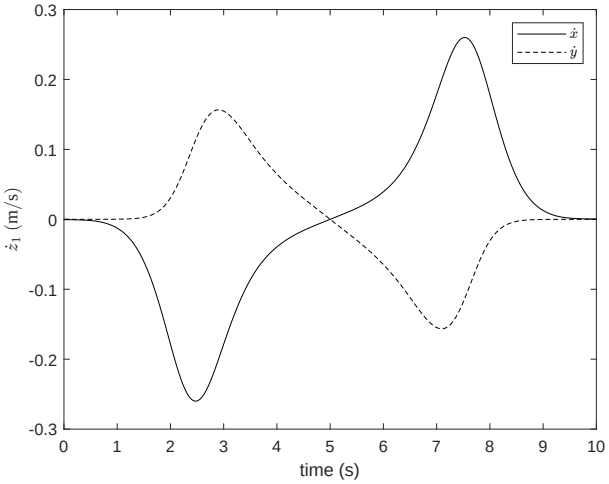
MUBOKAP Trace Path



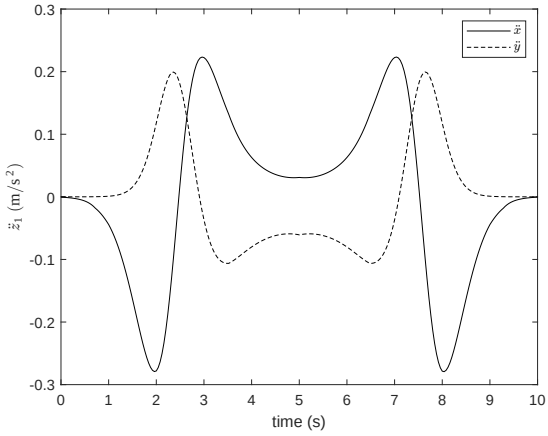
MUBOKAP x(t) and y(t), position time responses

By comparing the graphs, a discrepancy of approximately 1% can be observed between MATLAB and SolidWorks, at the time step 5 the error reaches its highest value. In the MUBOKAP results at the

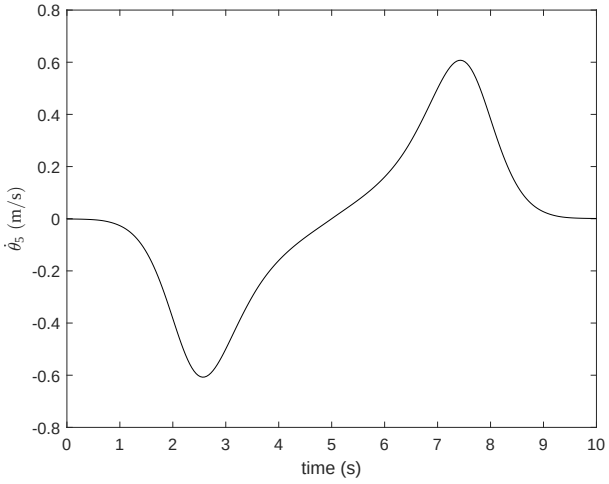
time step 5 its value of angular displacement is -1.09 rad. Comparing with SolidWorks (-1.163 rad) it gives an error of 6% and with MATLAB (-1.153 rad) an error of 5%. Once again due to the values in the MUBOKAP input file, which were poorly rounded to the hundredths place.



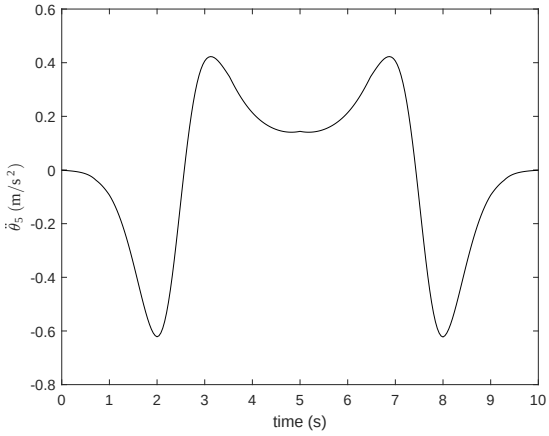
MUBOKAP OCD (wheel) velocity



MUBOKAP OCD (wheel) acceleration



MUBOKAP OCD (wheel) angular velocity



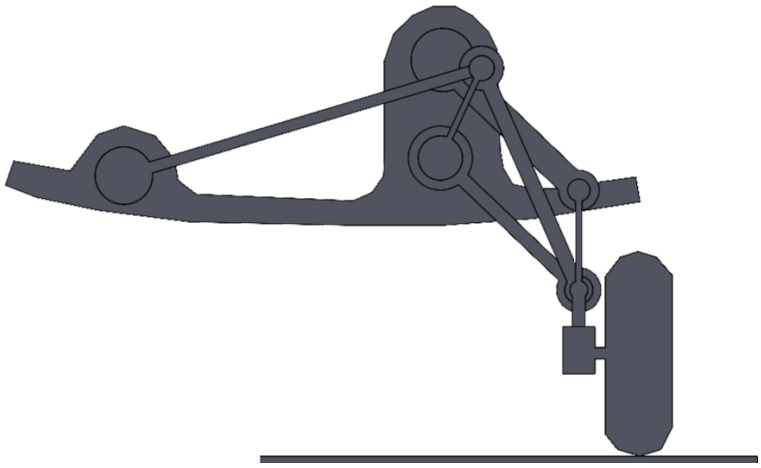
MUBOKAP OCD (wheel) angular acceleration

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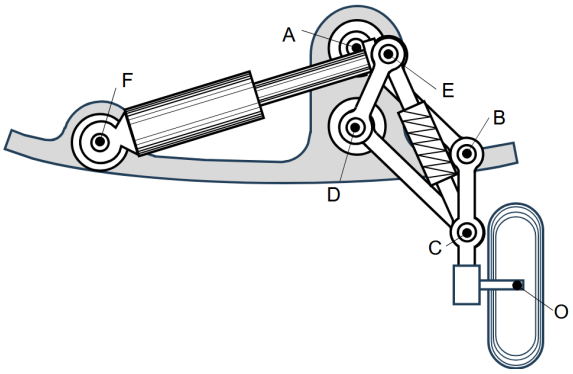
Dinâmica de Sistemas Mecânicos

Homework 4 – MUBODAP Landing Gear

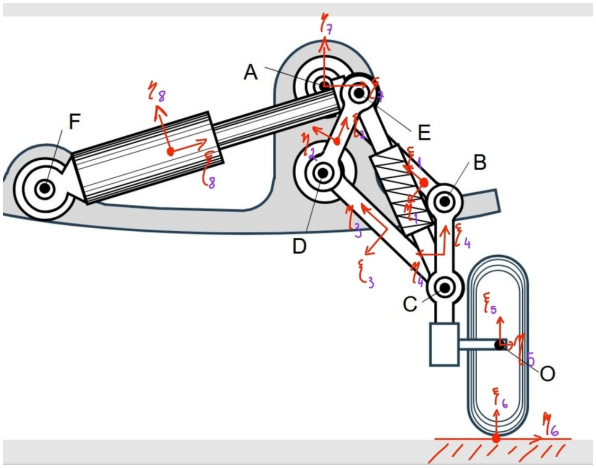
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1 Mechanism



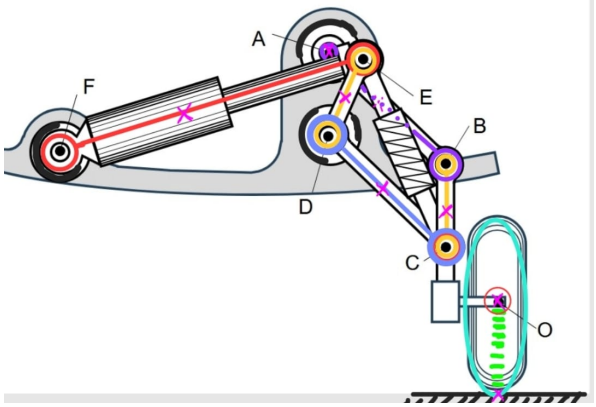
Landing gear, number 3



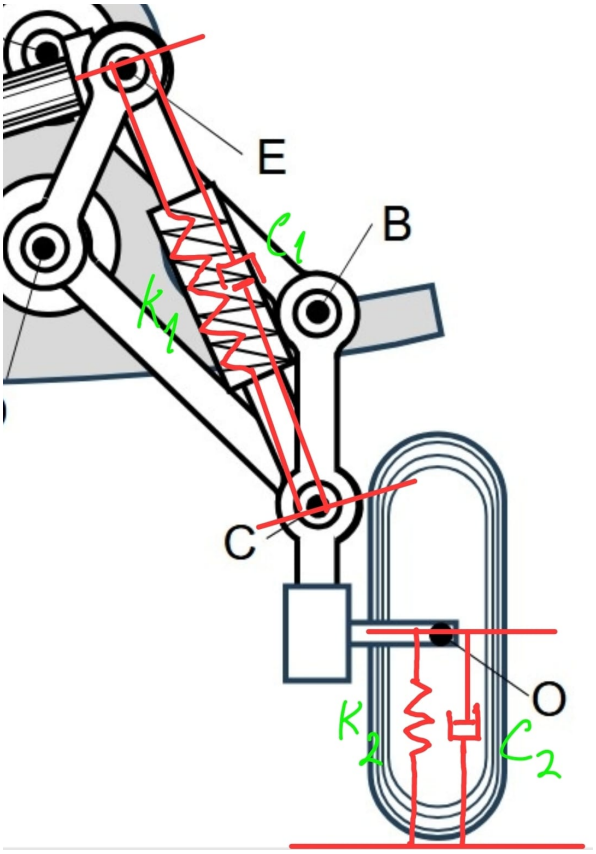
Landing gear with body numbering and local axis

Body	**Mass (kg)**	**Inertia (kg·m^2)**	**Connected to Body**	**Local Coordinates** (ξ, η)
1	3.500	0.30	7 4	$(0.13, 0.00)$ $(-0.43, 0.00)$
2	2.000	0.17	3 8	$(-0.04, 0.00)$ $(0.26, 0.00)$
3	3.000	1.19	7 4 2	$(0.22, 0.00)$ $(-0.35, 0.00)$ $(0.22, 0.00)$
4	2.000	0.94	1 3 5	$(0.14, 0.00)$ $(-0.16, 0.00)$ $(-0.16, 0.00)$
5	6.500	7.38	4 6 (Translação)	$(-0.16, 0.19)$ $(0.00, 0.00)$
6	0.000	0.000	5 (Translação)	$(0.00, -0.30)$
7	647.000	265.68	1 3 8	$(0.00, 0.00)$ $(0.00, -0.30)$ $(-0.96, -0.35)$
8	1.500	0.08	2 7	$(1.05, 0.00)$ $(-0.07, 0.00)$

The masses and moments of inertia were obtained using SolidWorks. The airplane frame (Body 7) has a mass of 647 kg, corresponding to the total weight of the aircraft minus one landing gear assembly, divided by three, as the aircraft is equipped with three landing gears.



Landing gear, connection planning. Green traced => Translation Joint, Rest => Revolute Joint



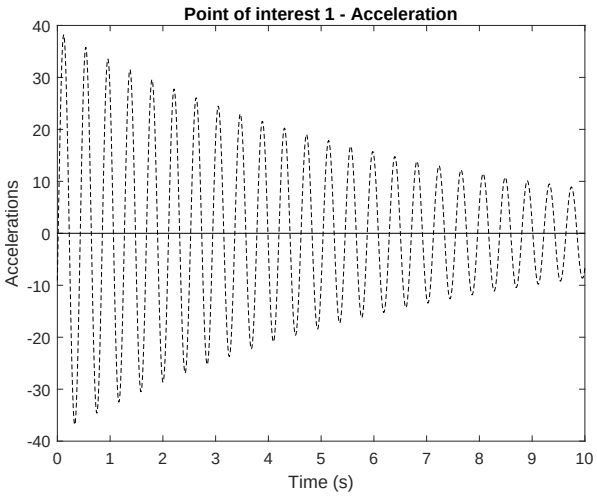
Landing gear Springs and Dampers

Body i	**Body** j	**Local Coord.** i	**Local Coord.** j	**Spring**	**Spring** L_0	**Damper**
5	6	$(0.00, 0.00)$	$(0.00, 0.00)$	$k_2 = 150.0kN/m$	0.30m	$c_2 = 200.0Ns/m$
2	4	$(0.26, 0.00)$	$(-0.16, 0.00)$	$k_1 = 280.0kN/m$	0.73m	$c_1 = 750.0kNs/m$

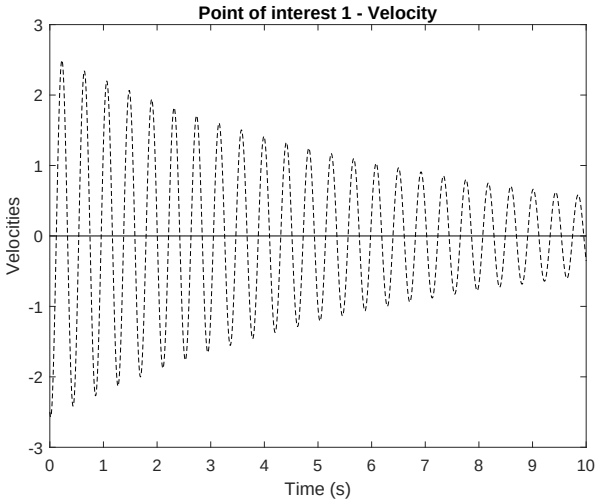
To prevent the aircraft frame from translating or rotating, it was constrained in the x and θ degrees of freedom. The ground (Body 6) was treated as a fixed body in the MUBODAP program, with all three degrees of freedom restrained.

2 Solution analysis

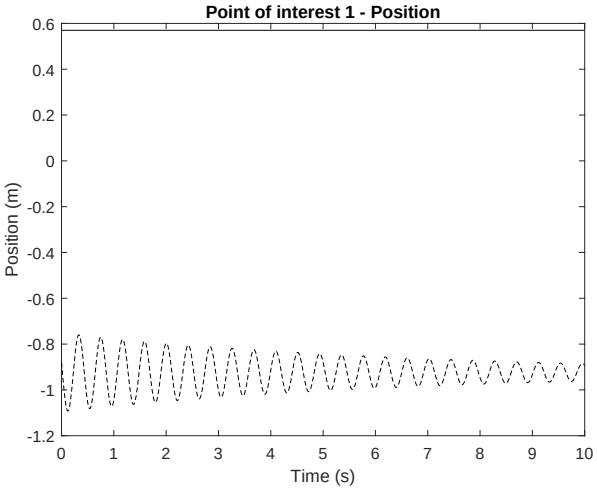
2.1 Wheel's center of mass



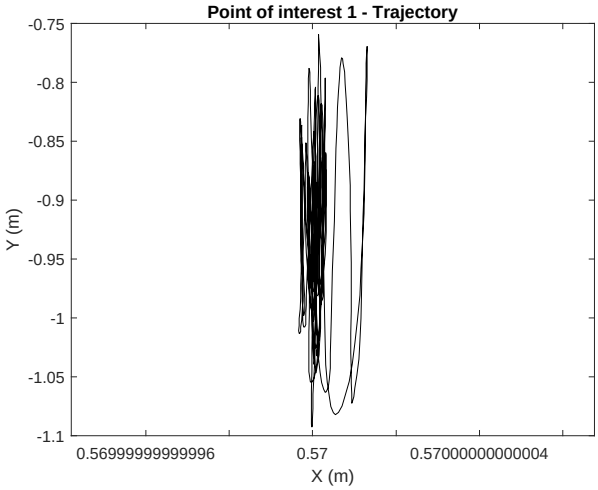
MATLAB – Wheel Accelerations (in x and y)



MATLAB – Wheel Velocities (in x and y)



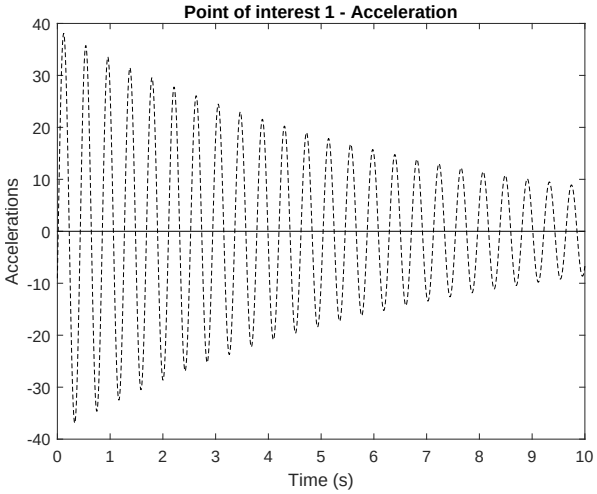
MATLAB – Wheel Positions (in x and y in time)



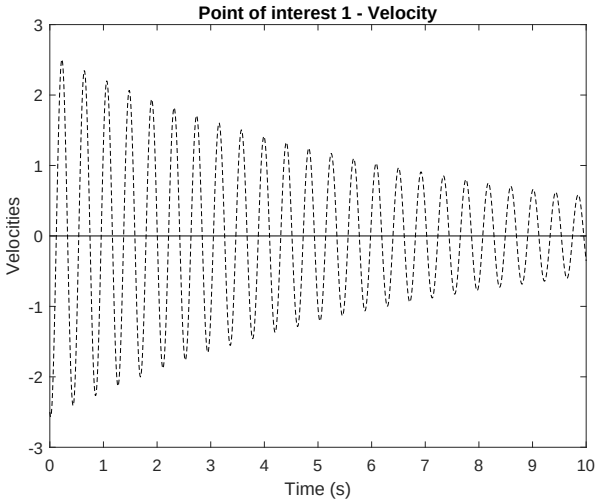
MATLAB – Wheel trajectory

The position of the wheel's center of mass does not exceed its radius, indicating that the model is performing correctly and that the wheel will not fail.

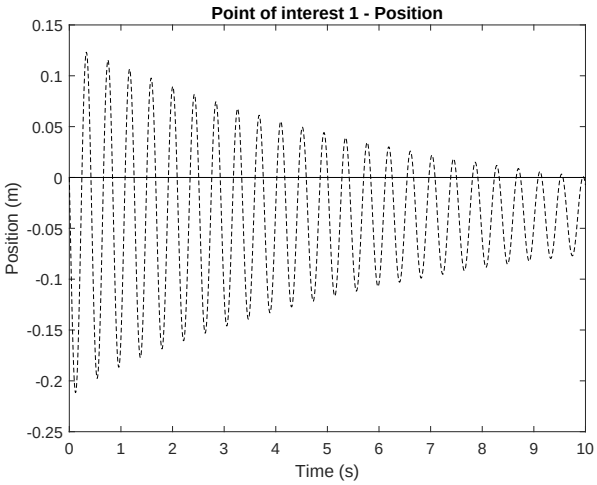
2.2 Airplane center of mass



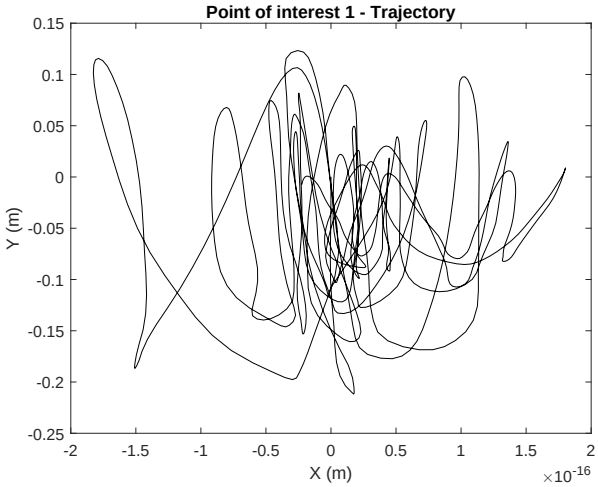
MATLAB – Airplane Frame Accelerations (in x and y)



MATLAB – Airplane Frame Velocities (in x and y)



MATLAB – Airplane Frame Positions (in x and y in time)



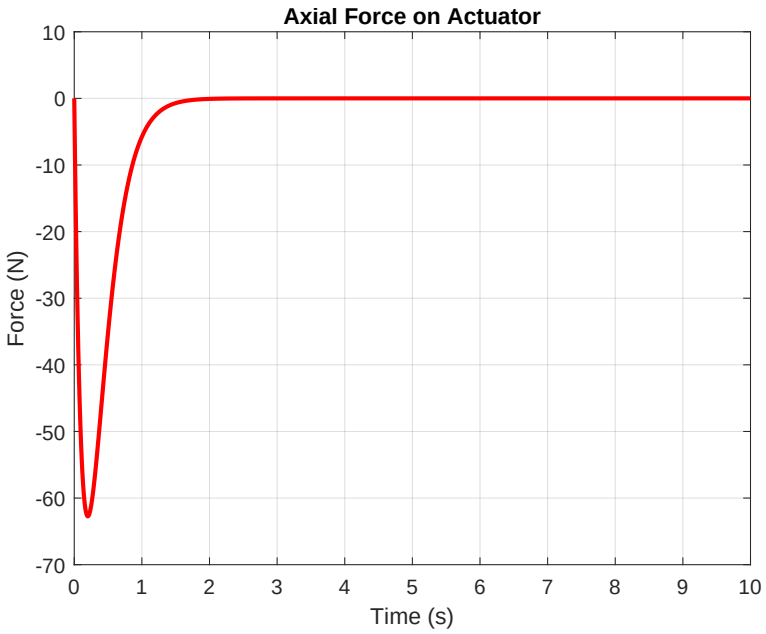
MATLAB – Airplane Frame trajectory

As intended, the aircraft does not exceed a displacement of 0.5 meters. The constants for k_1 and c_1 were completelly arbritary in order to achieve it.

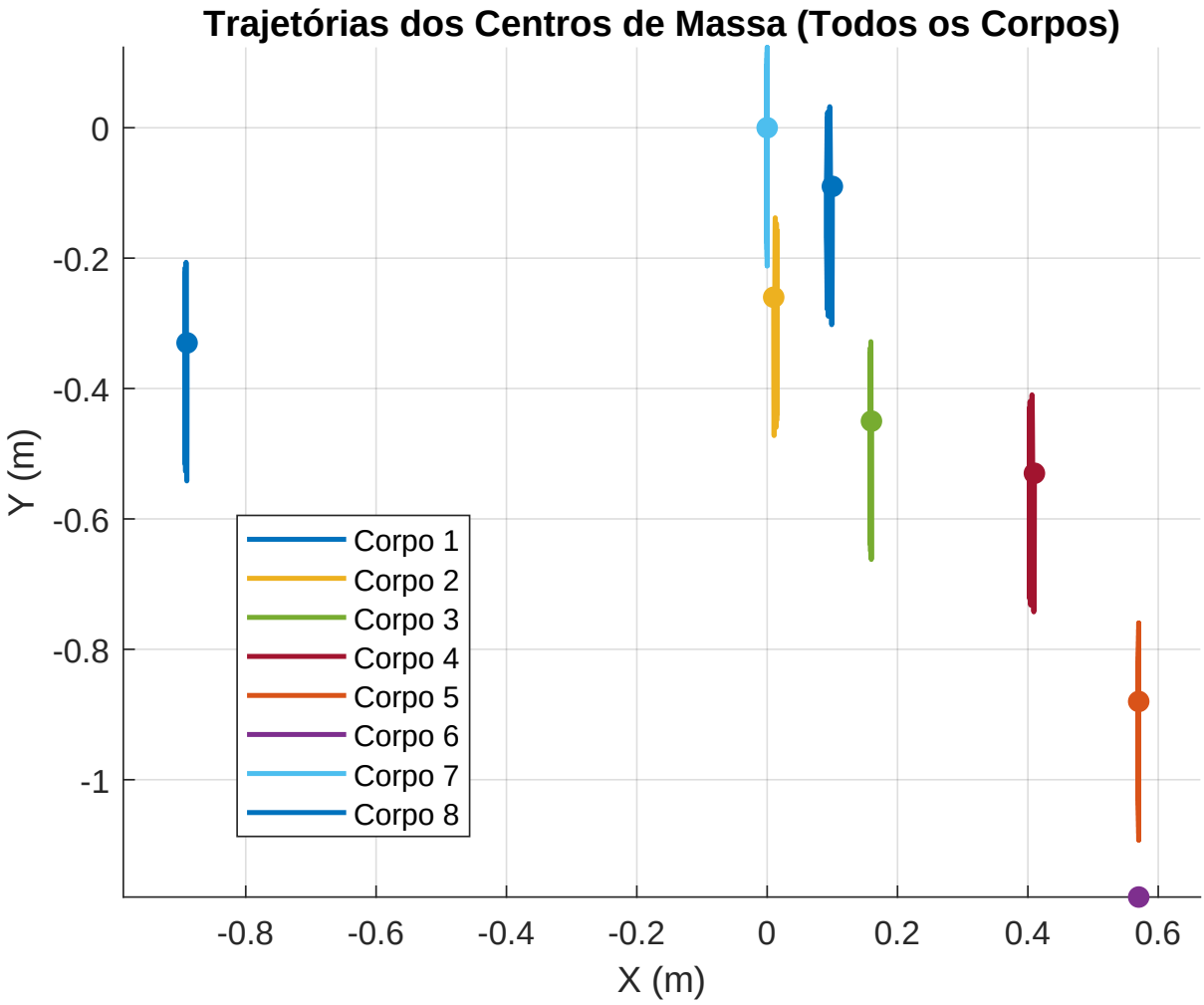
2.3 Axial Force from the pneumatic actuator

In order to obtain the axial force:

$$F_{\text{axial}} = k_1(L - L_0) + c_1\dot{L} \tag{1}$$



2.4 All bodies center of mass



MATLAB – Every body trajectory